Testing like an engineer

Programming · Level 4 · Lesson 4



In this lesson you will

- Write unit tests with assert, including edge cases.
- Find and fix a boundary bug in binary search.
- Work test-first, and review code with others.

Key ideas

- Test the edges: empty, one, first, last, missing.
- Red, green, refactor.
- Review the code, not the coder.

Activity: Break my code

1. Warm-up (3 min): Write search's edge cases on paper.
2. Trace (7 min): Trace the buggy search on [5].
3. Test (7 min): Write asserts first for a new function.
4. Review (3 min): Swap; one helpful comment each.

Tracing the boundary bug by hand is the moment it clicks: do it on paper before running anything.

Let’s discuss

1. Why did 90% of professional binary searches have bugs?
2. What does watching a test fail first prove?
3. How do you make code review feel safe?

Exit ticket

- Write three edge-case asserts for a function that returns the middle value of a list.